

ESP32 Smart Gas & Environment Safety Monitor

Complete Arduino / C++ Source Code

```
#include <Arduino.h>
#include <DHT.h>

// =====
// SMART GAS & ENVIRONMENT SAFETY MONITOR
// =====
//
// CONTROLLER:
// ESP32
//
// GAS SENSORS:
// MQ-2
// MQ-5
// MQ-7
// MQ-135
//
// ENVIRONMENT:
// DHT22
//
// OUTPUT:
// Serial Monitor
//
// ACTUATORS:
// Buzzer
// Relay 1
// Relay 2
//
// NOT INCLUDED:
// OLED
// PIR
// SIM/GSM
// =====

// =====
// PIN DEFINITIONS
// =====

// -----
// MQ GAS SENSORS
// -----

#define MQ5_PIN      32
#define MQ2_PIN      34
#define MQ135_PIN    33
#define MQ7_PIN      35

// -----
// BUZZER
// -----

#define BUZZER_PIN    2

// -----
// RELAYS
// -----

#define RELAY1_PIN    36
#define RELAY2_PIN    25

// -----
// DHT22
// -----

#define DHT_PIN        4
#define DHT_TYPE       DHT22

DHT dht(DHT_PIN, DHT_TYPE);

// =====
// TIMING
// =====
const unsigned long READ_INTERVAL = 3000;
```

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unsigned long previousMillis = 0;

// =====
// THRESHOLDS
// =====
//
// IMPORTANT:
// These are RAW ADC demonstration thresholds.
// They are NOT ppm values.
//
// Proper calibration is required before using this
// system for real gas-safety decisions.
// =====

const int MQ2_WARNING    = 2000;
const int MQ2_DANGER     = 2800;

const int MQ5_WARNING    = 2000;
const int MQ5_DANGER     = 2800;

const int MQ7_WARNING    = 1800;
const int MQ7_DANGER     = 2500;

const int MQ135_WARNING  = 2000;
const int MQ135_DANGER   = 2800;

// =====
// SYSTEM STATES
// =====

enum SystemState
{
    NORMAL,
    WARNING,
    DANGER
};

// =====
// SETUP
// =====

void setup()
{
    Serial.begin(115200);

    // Start DHT22
    dht.begin();

    // Buzzer
    pinMode(BUZZER_PIN, OUTPUT);
    digitalWrite(BUZZER_PIN, LOW);

    // Relays
    pinMode(RELAY1_PIN, OUTPUT);
    pinMode(RELAY2_PIN, OUTPUT);

    // Start relays OFF
    digitalWrite(RELAY1_PIN, LOW);
    digitalWrite(RELAY2_PIN, LOW);

    // =====
    // STARTUP INFORMATION
    // =====

    Serial.println();
    Serial.println("=====");
    Serial.println("      ESP32 SMART GAS SAFETY MONITOR");
    Serial.println("=====");

    Serial.println();

    Serial.println("PIN CONFIGURATION");
    Serial.println("-----");

    Serial.print("MQ-5      -> GPIO ");
    Serial.println(MQ5_PIN);

    Serial.print("MQ-2      -> GPIO ");
    Serial.println(MQ2_PIN);
    Serial.print("MQ-135    -> GPIO ");

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Serial.println(MQ135_PIN);

Serial.print("MQ-7    -> GPIO ");
Serial.println(MQ7_PIN);

Serial.print("DHT22    -> GPIO ");
Serial.println(DHT_PIN);

Serial.print("BUZZER    -> GPIO ");
Serial.println(BUZZER_PIN);

Serial.print("RELAY 1  -> GPIO ");
Serial.println(RELAY1_PIN);

Serial.print("RELAY 2  -> GPIO ");
Serial.println(RELAY2_PIN);

Serial.println();

Serial.println("NOT CONNECTED");
Serial.println("-----");
Serial.println("OLED");
Serial.println("PIR");
Serial.println("SIM/GSM");

Serial.println();

Serial.println("SYSTEM INITIALIZATION");
Serial.println("-----");

Serial.println("Starting sensors...");
Serial.println("MQ sensors require warm-up time.");
Serial.println();

delay(2000);

Serial.println("SYSTEM READY");
Serial.println();
}

// =====
// MAIN LOOP
// =====

void loop()
{
  if (millis() - previousMillis >= READ_INTERVAL)
  {
    previousMillis = millis();

    // =====
    // READ MQ SENSORS
    // =====

    int mq5Value  = analogRead(MQ5_PIN);
    int mq2Value  = analogRead(MQ2_PIN);
    int mq135Value = analogRead(MQ135_PIN);
    int mq7Value  = analogRead(MQ7_PIN);

    // =====
    // READ DHT22
    // =====

    float temperature = dht.readTemperature();
    float humidity    = dht.readHumidity();

    // =====
    // DISPLAY SENSOR DATA
    // =====

    Serial.println();
    Serial.println("=====");
    Serial.println("                SENSOR DATA");
    Serial.println("=====");

    Serial.print("MQ-5    : ");
    Serial.println(mq5Value);

    Serial.print("MQ-2    : ");
    Serial.println(mq2Value);

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Serial.print("MQ-135   : ");
Serial.println(mq135Value);

Serial.print("MQ-7     : ");
Serial.println(mq7Value);

// =====
// DHT22
// =====

Serial.println();

if (isnan(temperature) || isnan(humidity))
{
    Serial.println("DHT22 STATUS : SENSOR ERROR");
}
else
{
    Serial.print("Temperature : ");
    Serial.print(temperature);
    Serial.println(" °C");

    Serial.print("Humidity    : ");
    Serial.print(humidity);
    Serial.println(" %");
}

// =====
// SENSOR STATES
// =====

SystemState mq2State;
SystemState mq5State;
SystemState mq7State;
SystemState mq135State;

// =====
// MQ-2
// =====

if (mq2Value >= MQ2_DANGER)
{
    mq2State = DANGER;
}
else if (mq2Value >= MQ2_WARNING)
{
    mq2State = WARNING;
}
else
{
    mq2State = NORMAL;
}

// =====
// MQ-5
// =====

if (mq5Value >= MQ5_DANGER)
{
    mq5State = DANGER;
}
else if (mq5Value >= MQ5_WARNING)
{
    mq5State = WARNING;
}
else
{
    mq5State = NORMAL;
}

// =====
// MQ-7
// =====

if (mq7Value >= MQ7_DANGER)
{
    mq7State = DANGER;
}
else if (mq7Value >= MQ7_WARNING)

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```

{
    mq7State = WARNING;
}
else
{
    mq7State = NORMAL;
}

// =====
// MQ-135
// =====

if (mq135Value >= MQ135_DANGER)
{
    mq135State = DANGER;
}
else if (mq135Value >= MQ135_WARNING)
{
    mq135State = WARNING;
}
else
{
    mq135State = NORMAL;
}

// =====
// DISPLAY SENSOR STATUS
// =====

Serial.println();
Serial.println("-----");
Serial.println("                SENSOR STATUS");
Serial.println("-----");

Serial.print("MQ-5      : ");
printStats(mq5State);

Serial.print("MQ-2      : ");
printStats(mq2State);

Serial.print("MQ-135   : ");
printStats(mq135State);

Serial.print("MQ-7      : ");
printStats(mq7State);

// =====
// DETERMINE OVERALL CONDITION
// =====

bool dangerDetected =
    (mq2State == DANGER) ||
    (mq5State == DANGER) ||
    (mq7State == DANGER) ||
    (mq135State == DANGER);

bool warningDetected =
    (mq2State == WARNING) ||
    (mq5State == WARNING) ||
    (mq7State == WARNING) ||
    (mq135State == WARNING);

// =====
// DANGER
// =====

if (dangerDetected)
{
    digitalWrite(BUZZER_PIN, HIGH);

    // Relay actions
    digitalWrite(RELAY1_PIN, HIGH);
    digitalWrite(RELAY2_PIN, HIGH);

    Serial.println();
    Serial.println("=====");
    Serial.println("                !!! DANGER !!!");
    Serial.println("=====");
    Serial.println("HAZARDOUS GAS READING DETECTED.");
}

```

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Serial.println();

Serial.println("ACTION REQUIRED:");
Serial.println("1. EVACUATE THE AREA.");
Serial.println("2. DO NOT CREATE FLAMES OR SPARKS.");
Serial.println("3. MOVE TO A SAFE LOCATION.");
Serial.println("4. DO NOT INVESTIGATE THE GAS SOURCE.");
Serial.println("5. CONTACT THE RESPONSIBLE PERSON.");
Serial.println();

Serial.println("BUZZER : ON");
Serial.println("RELAY 1 : ON");
Serial.println("RELAY 2 : ON");
Serial.println("SYSTEM : EVACUATE");
}

// =====
// WARNING
// =====

else if (warningDetected)
{
    digitalWrite(BUZZER_PIN, HIGH);

    // Warning mode
    digitalWrite(RELAY1_PIN, LOW);
    digitalWrite(RELAY2_PIN, LOW);

    Serial.println();
    Serial.println("=====");
    Serial.println("                        WARNING");
    Serial.println("=====");

    Serial.println("ABNORMAL GAS SENSOR READING DETECTED.");
    Serial.println();

    Serial.println("RECOMMENDED ACTION:");
    Serial.println("1. CHECK THE AREA SAFELY.");
    Serial.println("2. IMPROVE VENTILATION.");
    Serial.println("3. MONITOR SENSOR VALUES.");
    Serial.println("4. CHECK FOR POSSIBLE GAS / SMOKE SOURCES.");
    Serial.println("5. PREPARE FOR EVACUATION IF LEVELS INCREASE.");
    Serial.println();

    Serial.println("BUZZER : ON");
    Serial.println("RELAY 1 : OFF");
    Serial.println("RELAY 2 : OFF");
    Serial.println("SYSTEM : WARNING");
}

// =====
// NORMAL
// =====

else
{
    digitalWrite(BUZZER_PIN, LOW);

    digitalWrite(RELAY1_PIN, LOW);
    digitalWrite(RELAY2_PIN, LOW);

    Serial.println();
    Serial.println("=====");
    Serial.println("                        SYSTEM NORMAL");
    Serial.println("=====");

    Serial.println("No abnormal gas reading detected.");
    Serial.println("Environment status : NORMAL");
    Serial.println("BUZZER : OFF");
    Serial.println("RELAY 1 : OFF");
    Serial.println("RELAY 2 : OFF");
    Serial.println("SYSTEM : MONITORING");
}

// =====
// TEMPERATURE CHECK
// =====

if (!isnan(temperature))
{
    Serial.println();

```

```

Serial.println("-----");
Serial.println("          TEMPERATURE STATUS");
Serial.println("-----");

if (temperature >= 40.0)
{
    Serial.println("WARNING: HIGH TEMPERATURE DETECTED.");
    Serial.println("ACTION : CHECK THE ENVIRONMENT.");
}
else
{
    Serial.println("Temperature status : NORMAL");
}
}

// =====
// END
// =====

Serial.println();
Serial.println("Next reading in 3 seconds...");
Serial.println("=====");
}

// =====
// PRINT SENSOR STATE
// =====

void printState(SystemState state)
{
    if (state == NORMAL)
    {
        Serial.println("NORMAL");
    }
    else if (state == WARNING)
    {
        Serial.println("WARNING");
    }
    else
    {
        Serial.println("DANGER");
    }
}

```